

Appendix 3

***Quality indicators for physiotherapy care process of patients with Whiplash-Associated Disorders (WAD):
steps of clinical reasoning, number of indicators, type of indicator, item measured, indicator, level of evidence* and
performance target***

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<i>Steps of clinical reasoning (number of indicators)</i>	<i>Type of indicator</i>	<i>Item</i>	<i>Indicator The average degree (in %) in which ...</i>	<i>Level of evidence*</i>	<i>Performance target</i>
I. Administration: 2 indicators (1-2)					
	Process Generic	Name, year of referral, referral, medical information	1. Patient's information is shared.	IV	≥80%
	Process Specific	Period since accident, request for help	2. Patient's request for help is noted.	IV	≥80%
II. History taking: 7 indicators (3 – 9)					
Ila. Sociodemographic characteristics	Process Generic	Ila. age, gender, educational level, family status, employment status	3. Patients were subjected to a methodically performed history taking, and sociodemographic characteristics are noted.	IV	≥80%
IIb. Accident related information	Process Specific	IIb. location in vehicle, use of seatbelt, use of positioned headrest, anticipated collision, type of trauma, time of onset of whiplash- related complaints	4. Patients were subjected to a methodically performed history taking, and accident related information is noted.	IV	≥80%
IIc. Pre-existent functioning and health status	Process Generic	IIc. Pre-existent activity limitations, participation problems, job-related problems	5. Patients were subjected to a methodically performed history taking, and pre-existent functioning is noted.	IV	≤30%

	Process Specific	IIc. Previous history of neck injury, pre-existent neck pain and/or stiffness, and/or irradiating arm pain, pre-existent pain else, comorbidity, relevant medication use	6. Patients were subjected to a methodically performed history taking, and pre-existent health status is noted.	IV	≤30%
IIId. Previous diagnostics and treatment	Process Specific	IIId. Previous medical imaging neck diagnostics, cervical soft collar after trauma, pain medication, modalities of (manual) physiotherapy, recovery after previous treatment	7. Patients were subjected to a methodically performed history taking, and previous diagnostics and treatment are noted.	IV	≤30%
IIe. Current health status and recovery rate since accident	Process Generic	IIe. Impairments in musculoskeletal neck functions, activity limitations, participation problems, job-related problems	8. Patients were subjected to a methodically performed history taking, and current functioning are noted.	IV	≥80%
	Process Specific	IIe. Recovery rate since accident, type and number of complaints, type of signs and symptoms, inventory prognostic factors, pain medication, *symptoms related to the presence of central sensitization (*since 2009)	9. Patients were subjected to a methodically performed history taking, and recovery rate since accident, prognostic factors and the presence of central sensitization are asked and administrated.	IV	≥80%
III. Objectives of examination; 1 indicator (10)					
IIIa Objectives of musculoskeletal examination	Process Specific	Examination objectives in agreement with patient's history taking and supplementary medical data, choice of clinical	10. Examination objectives in agreement with patient's history are noted, and choice of	IV	≥80%

IIIb. Objectives of neurological examination		musculoskeletal, neurological and *oto-neurological tests, and *selection of psychological questionnaires (*since 2000)	clinical tests and psychological questionnaires is noted.		
IIIc. Objectives of oto-neurological examination					
IIId. Objectives of psychological examination					
IV. Clinical examination; 4 indicators (11-14)					
IVa Musculoskeletal examination	Process Specific	Cervical testing (observation of posture, range of motion and palpation) in agreement with objectives of musculoskeletal examination	11. The results of clinical evaluation of cervical musculoskeletal functions testing are noted.	II - IV	≥80%
IVb. Neurological examination	Process Specific	Testing of sensory functions and pain, muscles functions, reflexes and coordination, and testing of cranial nerve functions (partly incorporated in oto-neurological examination, particularly trigeminal nerve) in agreement with objectives of neurological examination	12. The results of clinical evaluation of neurological functions are noted.	IV	≥80%
IVc. Oto-neurological examination	Process Specific	Standing and gait testing, dizziness test, positional testing, eyes movement test in agreement with objectives of *oto-neurological examination (*since 2000)	13. The results of clinical evaluation of equilibrium and dizziness / vertigo are noted.	IV	≥80%
IVd. Psychological examination	Process Specific	Observation of pain behavior, and *questionnaires (Fear- Avoidance Beliefs Questionnaire – FABQ – and Pain Coping Inventory -PCI) (*since 2002)	14. The results of examination of psychological functions and tests are noted.	II - IV	≥80%
V. Analysis and conclusion of diagnostic process: 2 indicators (15,16)					

	Process Specific	Classification Whiplash-Associated Disorders, time phase since accident, recovery in time since accident, determination of health profile A / B / C, prognostic factors, use of questionnaires, referral to GP in case if insufficient or no results expected, indication physiotherapy	15. Individual health profile addressed to the whiplash injury since accident, an indication of treatment prognosis, and an indication for physiotherapy have been established and are noted.	II- IV	≥80%
	Process Specific	*Presence of central sensitization (*since 2009),	16. Presence of central sensitization is noted.	IV	Non-defined
VI. Treatment plan: 3 indicators (17-19)					
	Process Specific	Main treatment goals in different time phases since accident and in agreement with individual health profile, prognostic duration of treatment period and prognostic number of treatment sessions, *pre-treatment measures pain (VAS) and functioning (NDI) (*since 2002), treatment plan in agreement with patient	17. Treatment goals are methodically determined and noted in agreement with individual prognostic health profile, time phase since accident, and with patient.	II-IV	≥80%
	Process Generic		18. Prognostic treatment period and number of treatment sessions are noted.	IV	≥80%
	Process Generic		19. Pre-treatment scores VAS and NDI are measured and noted.	I	≥80%
VII. Treatment: 2 indicators (20 ,21)					

	Process Specific	Physiotherapy modalities with best available evidence in different time phases since accident in agreement with patient profile and treatment goals, and check for side effects	20. Physiotherapy modalities in agreement with treatment goals in time phases since accident and health profile, and with best available evidence are applied and noted. 21. Treatment effects and side effects are noted in patient's record.	II-IV IV	≥80% ≥80%
VIII. Evaluation; 5 indicators (22-26)					
VIIIa. Evaluation during treatment	Process Specific	Perceived result per treatment goal, regular and systematic evaluation and, if necessary, adjustment of treatment goals and treatment modalities, contact physician if insufficient treatment result	22. A methodically performed evaluation of treatment goals and treatment modalities are noted.	IV	≥80%
VIIIb. Final evaluation	Outcome Generic	Final subjective and objective evaluation of treatment goals, *post-treatment measures (pain (VAS) and functioning (NDI) (since 2002), *global perceived effect (GPE) (*since 2002), return to work, duration of treatment period and number of treatment sessions at the end of total treatment	23. Reached treatment goals and returned to work are subjectively evaluated and noted. 24. Post-treatment scores (pain (VAS) and functioning (NDI)) are measured and noted. 25. Global perceived effect is measured and noted. 26. Duration of treatment period and number of treatment sessions are noted.	IV I II IV	≥80% ≥80% ≥80% ≥80%

IX. Discharge: 2 indicators (27, 28)					
	Process Generic	Reason for discharge, written report to physician in copy to patient	27. A final report is written and noted.	IV	≥80%
	Process Specific	If necessary, arrangement of aftercare	28. Aftercare is arranged	IV	Non-defined

* Levels of evidence: I = systematic review or >2 high-quality controlled trials or high-quality diagnostic studies or high-quality psychometric studies; II = two high quality controlled trials or high-quality diagnostic studies or high-quality psychometric studies; III: high quality non-controlled trials or low-quality diagnostic studies or low-quality psychometric studies; IV: experts opinion and professional consensus or standard.